

Package ‘rpbnb’

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Type Package

Title Random-Parameter Bivariate Negative Binomial Regression

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Description Maximum-likelihood estimation of bivariate negative binomial (NB2) regression models with Famoye/Sarmanov or discrete-copula (Frank, Gaussian, or Clayton) dependence, and maximum-simulated-likelihood estimation of a bivariate random-parameter negative binomial model under either dependence structure, with a random-coefficient data simulator, diagnostics, and standard model methods.

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bnb_elasticities	<i>Elasticities and semi-elasticities for a bivariate NB model</i>
------------------	--

Description

For continuous regressors, the elasticity is $\beta_j E[x_j]$; for binary (0/1) regressors, the semi-elasticity is $\exp(\beta_j) - 1$ (proportional change moving from 0 to 1).

Usage

```
bnb_elasticities(
  fit,
  which = c("y1", "y2", "both"),
  type = c("AME", "MEM"),
  vars = NULL,
  include_intercept = FALSE,
  digits = 4,
  print_output = TRUE
)
```

Arguments

fit	A bnb_fit object from fit_bnb() .
which	Which margin(s): "y1", "y2", or "both".
type	"AME" (uses sample mean of x) or "MEM" (uses mean design row).
vars	Optional variable names to restrict output.
include_intercept	Logical; include the intercept term.

digits Number of decimal places for printed output.
 print_output Logical; if FALSE, suppress printing.

Value

A data frame (single margin) or a named list of data frames (both), each with columns Name, Estimate, StdErr, z, p, Signif, var_type.

Examples

```
d <- read.csv(system.file("extdata", "rwm1984_clean.csv", package = "rpbnb"))
fit <- fit_bnb(docvis ~ outwork + age, hospvis ~ outwork, data = d,
               dependence = "famoye")
bnb_elasticities(fit, which = "both", type = "AME")
```

bnb_gof	<i>Goodness of fit for a bivariate NB model</i>
---------	---

Description

Reports log-likelihood, AIC, BIC, and four pseudo-R-squared measures (McFadden, McFadden adjusted, Cox-Snell, Nagelkerke) relative to an intercept-only null model refit with the same dependence structure (copula fits rebuild the copula from `fit$cop_family`). Pseudo-R-squared values are returned raw and are not clamped to $[0, 1]$; a negative value flags a full model that fits worse than the null.

Usage

```
bnb_gof(fit, digits = 4, print_output = TRUE)
```

Arguments

fit A bnb_fit object from `fit_bnb()`.
 digits Number of decimal places for printed output.
 print_output Logical; if FALSE, suppress printing and return the result invisibly.

Value

Invisibly, a list with n, k, logLik_full, logLik_null, AIC, BIC, a named numeric vector pseudoR2, and the null_fit.

Examples

```
d <- read.csv(system.file("extdata", "rwm1984_clean.csv", package = "rpbnb"))
fit <- fit_bnb(docvis ~ outwork + age, hospvis ~ outwork, data = d,
               dependence = "famoye")
bnb_gof(fit)
```

bnb_marginal_effects *Marginal effects for a bivariate NB model*

Description

Average marginal effects (AME) or marginal effects at the mean (MEM) for each margin. Continuous and binary (0/1) regressors are auto-detected; continuous effects use $\partial E[Y]/\partial x_j = \beta_j \mu$ and binary effects use $E[Y|x_j = 1] - E[Y|x_j = 0]$.

Usage

```
bnb_marginal_effects(
  fit,
  which = c("y1", "y2", "both", "all"),
  type = c("AME", "MEM"),
  vars = NULL,
  include_intercept = FALSE,
  digits = 4,
  print_output = TRUE
)
```

Arguments

<code>fit</code>	A <code>bnb_fit</code> object from <code>fit_bnb()</code> .
<code>which</code>	Which margin(s): "y1", "y2", "both", or "all".
<code>type</code>	"AME" (average marginal effect) or "MEM" (effect at the mean).
<code>vars</code>	Optional variable names or indices to restrict output.
<code>include_intercept</code>	Logical; include the intercept term.
<code>digits</code>	Number of decimal places for printed output.
<code>print_output</code>	Logical; if FALSE, suppress printing.

Value

A data frame (single margin) or a named list of data frames (both), each with columns Name, Estimate, StdErr, z, p, Signif, var_type.

Examples

```
d <- read.csv(system.file("extdata", "rwm1984_clean.csv", package = "rpbnb"))
fit <- fit_bnb(docvis ~ outwork + age, hospvis ~ outwork, data = d,
  dependence = "famoye")
bnb_marginal_effects(fit, which = "y1", type = "AME")
```

bnb_residual_checks	<i>Residual checks for a bivariate NB model</i>
---------------------	---

Description

Formal residual diagnostics for a `fit_bnb()` or `fit_rpbnb()` fit: a normality test on the randomized quantile residuals (Shapiro-Wilk for $n \leq 5000$, else Kolmogorov-Smirnov vs $N(0,1)$); the NB2 dispersion statistic ($\text{sum}(\text{pearson}^2) / (n - k)$) per margin; the cross-margin RQR correlation (a check that the Famoye/copula dependence captured the association); an outlier count/index list ($|\text{RQR}| > \text{outlier_z}$); and a composite misspecification verdict combining these signals.

Usage

```
bnb_residual_checks(
  fit,
  seed = NULL,
  outlier_z = 3,
  digits = 4,
  print_output = TRUE
)
```

Arguments

<code>fit</code>	A <code>bnb_fit</code> or <code>rpbnb_fit</code> object.
<code>seed</code>	Optional integer seed for the RQR randomization.
<code>outlier_z</code>	Absolute-RQR threshold flagging an outlier (default 3).
<code>digits</code>	Decimal places for printed output.
<code>print_output</code>	Logical; if FALSE, suppress printing.

Value

Invisibly, an object of class `bnb_residual_checks`.

<code>copula</code>	<i>Specify a copula dependence structure</i>
---------------------	--

Description

Pass the result as the dependence argument to `fit_bnb()` or `fit_rpbnb()` to estimate a discrete-copula bivariate NB model instead of the default Famoye/Sarmanov dependence, or as the copula argument to `simulate_rpbnb_copula()` to simulate from one. The joint pmf is built from the two NB2 marginal CDFs and the chosen copula CDF (rectangle differencing); see the package vignette for the "Copula dependence" section.

Usage

```
copula(family = c("frank", "normal", "kimeldorf"), par = NULL)
```

Arguments

family	One of "frank", "normal" (Gaussian), or "kimeldorf" (Clayton).
par	Optional native dependence parameter (Frank's theta, the Gaussian rho, or Clayton's theta) used by <code>simulate_rpbnb_copula()</code> to generate data. Ignored by <code>fit_bnb()</code> and <code>fit_rpbnb()</code> , which estimate the parameter from the data.

Value

An object of class `rpbnb_copula`.

Examples

```
copula("frank")
copula("normal", par = 0.3)
copula("kimeldorf")
```

fit_bnb	<i>Fit a bivariate negative binomial regression model</i>
---------	---

Description

Fit a bivariate negative binomial regression model

Usage

```
fit_bnb(
  formula_1,
  formula_2,
  data,
  dependence = c("independence", "famoye"),
  start = NULL,
  control = rpbnb_control()
)
```

Arguments

formula_1, formula_2	Model formulas for the two count outcomes.
data	A data frame.
dependence	Dependence structure: "independence" (two univariate NB2 margins), "famoye" (Famoye/Sarmanov bivariate NB), or a <code>copula()</code> object (Frank / Gaussian / Clayton discrete-copula bivariate NB; the dependence parameter is estimated).
start	Optional starting parameter vector. May be positional (length equal to the number of parameters) or named; a named vector is reordered to the canonical parameter order and a named partial vector is merged into the defaults (unknown or duplicate names are rejected). When <code>start</code> is <code>NULL</code> , the famoye path uses a multi-start policy: it optimizes from both an all-zero mean-coefficient start and marginal <code>glm.nb</code> starts and keeps the better converged objective (the frozen-bounds gradient makes the objective start-sensitive and neither start dominates).
control	An <code>rpbnb_control()</code> object. The famoye and copula estimators both use BFGS, the only optimizer <code>control\$method</code> accepts.

Value

An object of class `bnb_fit`.

Examples

```
d <- read.csv(system.file("extdata", "rwm1984_clean.csv", package = "rpbnb"))
fit <- fit_bnb(docvis ~ outwork, hospvis ~ outwork, data = d,
              dependence = "famoye")
summary(fit)

# Gaussian copula dependence instead of Famoye/Sarmanov
fit_cop <- fit_bnb(docvis ~ outwork, hospvis ~ outwork, data = d,
                  dependence = copula("normal"))
fit_cop$cop_tau # estimated Kendall's tau
```

fit_rpbnb	<i>Fit a bivariate random-parameter negative binomial model</i>
-----------	---

Description

Maximum simulated likelihood estimation with normal random coefficients and Famoye/Sarmanov dependence. Random coefficients are selected per equation by name via `random_1` / `random_2`.

Usage

```
fit_rpbnb(
  formula_1,
  formula_2,
  data,
  random_1 = NULL,
  random_2 = NULL,
  draws = 400,
  draw_type = "halton",
  seed = 1234,
  start = NULL,
  control = rpbnb_control(),
  dependence = "famoye"
)
```

Arguments

<code>formula_1, formula_2</code>	Model formulas for the two count outcomes.
<code>data</code>	A data frame.
<code>random_1, random_2</code>	Random coefficients per equation. Either a character vector of <code>model.matrix</code> column names (all Normal), or a named list whose values are a distribution name ("normal", "lognormal", "uniform", "triangular") or a list <code>list(dist = ..., sign = ...)</code> (sign is -1/1 and lognormal-only). NULL means all-fixed for that equation.
<code>draws</code>	Number of simulation draws for the optimization.

draw_type	Quasi-random draw type. Only "halton" is supported in this version.
seed	Random seed for the simulation draws.
start	Optional starting parameter vector.
control	An <code>rpbnb_control()</code> object. Estimation uses BFGS, the only optimizer <code>control\$method</code> accepts.
dependence	Dependence structure: "famoye" (default; Famoye/Sarmanov) or an <code>copula()</code> object for copula dependence (Frank / Gaussian / Clayton). Both paths use the multithreaded (OpenMP) C++ simulated likelihood; the copula path is more numerically expensive per evaluation (discrete-copula pmf + per-draw NB CDF corners), so fits typically take noticeably longer than the Famoye path at comparable draws/n. Random coefficients on 0/1 dummy regressors are weakly identified under the copula path (NB dispersion trades off against the random-coefficient scale); prefer random coefficients on continuous regressors.

Value

An object of class `rpbnb_fit`.

Examples

```
sim <- simulate_rpbnnb(n = 600,
  beta1 = c("(Intercept)" = 0.2, x1 = 0.4),
  beta2 = c("(Intercept)" = 0.1, x1 = -0.3),
  random_1 = list(x1 = list(sd = 0.5)),
  dispersion = c(m1 = 0.4, m2 = 0.5), seed = 1)
fit <- fit_rpbnnb(y1 ~ x1, y2 ~ x1, data = sim$data, random_1 = "x1",
  draws = 100, control = rpbnb_control(compute_se = FALSE))
coef(fit)

# Copula dependence instead of Famoye/Sarmanov (slower; fewer draws here
# for a quick example -- use more in practice)
sim_cop <- simulate_rpbnnb_copula(n = 600,
  beta1 = c("(Intercept)" = 0.2, x1 = 0.4),
  beta2 = c("(Intercept)" = 0.1, x1 = -0.3),
  random_1 = list(x1 = list(sd = 0.3)),
  dispersion = c(m1 = 0.4, m2 = 0.5),
  copula = copula("normal", par = 0.5), seed = 1)
fit_cop <- fit_rpbnnb(y1 ~ x1, y2 ~ x1, data = sim_cop$data, random_1 = "x1",
  dependence = copula("normal"), draws = 100,
  control = rpbnb_control(compute_se = FALSE))
tanh(coef(fit_cop)[["z_theta"]]) # estimated copula rho
```

plot.bnb_fit

Residual diagnostic plots for a bivariate NB model

Description

Four base-graphics panels per margin: residuals-vs-fitted, a normal QQ plot of the randomized quantile residuals, a histogram of the RQR with an $N(0,1)$ overlay, and a scale-location plot. The QQ and histogram panels always use RQR (only these are approximately $N(0,1)$ under a correct count model).

Usage

```
## S3 method for class 'bnb_fit'
plot(
  x,
  margin = c("both", "y1", "y2"),
  which = 1:4,
  resid_type = "quantile",
  seed = NULL,
  ...
)
```

Arguments

x	A bnb_fit object from <code>fit_bnb()</code> .
margin	Which margin to plot: "both" (default), "y1", or "y2".
which	Integer subset of panels 1:4 (1 = residuals-vs-fitted, 2 = QQ, 3 = histogram, 4 = scale-location).
resid_type	Residual type for panels 1 and 4: "quantile" (default), "pearson", "deviance", or "response".
seed	Optional integer seed for the RQR randomization.
...	Unused.

Value

NULL, invisibly (called for the side effect of drawing).

plot.rpbnb_fit	<i>Residual diagnostic plots for a random-parameter bivariate NB model</i>
----------------	--

Description

Four base-graphics panels per margin, as for `plot.bnb_fit()`, built on the mixture-based randomized quantile residuals. `resid_type = "deviance"` is not available for `rpbnb_fit`.

Usage

```
## S3 method for class 'rpbnb_fit'
plot(
  x,
  margin = c("both", "y1", "y2"),
  which = 1:4,
  resid_type = "quantile",
  seed = NULL,
  ...
)
```

Arguments

<code>x</code>	An <code>rpbnb_fit</code> object from <code>fit_rpbnb()</code> .
<code>margin</code>	Which margin to plot: "both" (default), "y1", or "y2".
<code>which</code>	Integer subset of panels 1:4.
<code>resid_type</code>	Residual type for panels 1 and 4: "quantile" (default), "pearson", or "response".
<code>seed</code>	Optional integer seed for the RQR randomization.
<code>...</code>	Unused.

Value

NULL, invisibly.

<code>residuals.bnb_fit</code>	<i>Residuals for a bivariate NB model</i>
--------------------------------	---

Description

Per-margin residuals for a fixed-coefficient `fit_bnb()` model. Each margin is NB2 with fitted mean μ and dispersion m (size = $1/m$). "quantile" returns randomized quantile residuals (Dunn & Smyth 1996), which are approximately $N(0,1)$ under a correct model and are the recommended residual for normality-style diagnostics on count data.

Usage

```
## S3 method for class 'bnb_fit'
residuals(
  object,
  type = c("quantile", "pearson", "deviance", "response"),
  margin = c("both", "y1", "y2"),
  seed = NULL,
  ...
)
```

Arguments

<code>object</code>	A <code>bnb_fit</code> object from <code>fit_bnb()</code> .
<code>type</code>	Residual type: "quantile" (default), "pearson", "deviance", or "response".
<code>margin</code>	Which margin: "both" (default), "y1", or "y2".
<code>seed</code>	Optional integer seed for the quantile-residual randomization (ignored for other types); does not disturb the caller's RNG stream.
<code>...</code>	Unused.

Value

A numeric vector for a single margin, or a two-column data frame (y1, y2) for `margin = "both"`.

residuals.rpbnb_fit	<i>Residuals for a random-parameter bivariate NB model</i>
---------------------	--

Description

Per-margin residuals for an `fit_rpbnb()` model. Each margin is a mixture of NB2 distributions over the random-coefficient draws. "quantile" returns randomized quantile residuals (Dunn & Smyth 1996) from the exact mixture predictive CDF (the recommended residual); "pearson" uses the exact mixture marginal variance. "deviance" is not defined for the mixture and errors.

Usage

```
## S3 method for class 'rpbnb_fit'
residuals(
  object,
  type = c("quantile", "pearson", "deviance", "response"),
  margin = c("both", "y1", "y2"),
  seed = NULL,
  ...
)
```

Arguments

object	An rpbnb_fit object from <code>fit_rpbnb()</code> .
type	Residual type: "quantile" (default), "pearson", or "response". "deviance" is not supported for rpbnb_fit.
margin	Which margin: "both" (default), "y1", or "y2".
seed	Optional integer seed for the quantile-residual randomization; does not disturb the caller's RNG stream.
...	Unused.

Value

A numeric vector for a single margin, or a two-column data frame (y1, y2) for margin = "both".

rpbnb_control	<i>Control parameters for rpbnb estimators</i>
---------------	--

Description

Control parameters for rpbnb estimators

Usage

```
rpbnb_control(
  method = c("BFGS"),
  iterlim = 300L,
  reltol = 1e-08,
  print_level = 2L,
  draws_hessian = 100L,
  halton_burn = 300L,
  n_cores = 1L,
  compute_se = TRUE,
  hessian = c("numeric", "analytic"),
  se_method = c("numeric", "opg", "analytic"),
  hess_eps = 1e-05,
  hess_r = 4L
)
```

Arguments

method	Optimizer used by the fitters. Only "BFGS" is implemented and wired through; it is the sole accepted value.
iterlim	Maximum optimizer iterations.
reltol	Relative convergence tolerance.
print_level	Verbosity passed to the optimizer (0 = silent, default = 2 for detailed progress).
draws_hessian	Retained for backward compatibility but currently unused: the random-parameter numeric Hessian is now taken with the same optimization draws that produced the estimate (same-draw curvature), so it no longer resimulates a separate Hessian draw set. Ignored by <code>fit_bnb()</code> .
halton_burn	Number of leading Halton points discarded before forming the simulation draws.
n_cores	Worker processes for the optional cluster path (1 = sequential).
compute_se	If FALSE, skip the Hessian and standard errors.
hessian	How <code>fit_bnb()</code> (famoye) computes the Hessian for standard errors: "numeric" (default, <code>numDeriv::hessian()</code>) or "analytic" (the closed-form Famoye (2010) Appendix Hessian). Both freeze the lambda-bounds at the optimum and yield the same observed-information SEs.
se_method	Standard-error method for <code>fit_rpbnb()</code> (the random-parameter model): "numeric" (default) uses the <code>numDeriv::hessian()</code> observed- information Hessian; "analytic" uses the closed-form observed-information Hessian (Famoye (2010) per-draw second derivatives assembled via the Louis mixture formula) – exact and much faster than "numeric" for larger models; "opg" uses the BHHH / outer-product-of-gradients information from the per-observation scores – fastest, but relies on the information-matrix equality so it is unreliable for parameters at a boundary (e.g. a random- coefficient SD estimated near 0). For copula dependence (<code>fit_rpbnb()</code> with <code>dependence = copula(...)</code>), only "opg" (recommended) and "numeric" are available; "analytic" is not implemented for the copula path and errors. Ignored by <code>fit_bnb()</code> .
hess_eps, hess_r	Step and Richardson order for <code>numDeriv::hessian()</code> (used only when <code>hessian = "numeric"</code>).

Value

An object of class `rpbnb_control` (a named list).

Examples

```
rpbnb_control(method = "BFGS", iterlim = 200)
rpbnb_control(hessian = "analytic")
```

rpbnb_elasticities	<i>Elasticities and semi-elasticities for a random-parameter bivariate NB model</i>
--------------------	---

Description

Continuous elasticities $x_{ij} (\partial \mu_i / \partial x_{ij}) / \mu_i$ and binary semi-elasticities $\mu_i(x_j = 1) / \mu_i(x_j = 0) - 1$, built on the Monte-Carlo integrated population mean $\mu_i = E_\beta[\exp(x'_i \beta)]$ of an `fit_rpbnb()` fit. Under fixed coefficients these reduce to $\beta_j E[x_j]$ and $\exp(\beta_j) - 1$. Standard errors use a numeric delta method over the equation's mean and log-scale parameters.

Usage

```
rpbnb_elasticities(
  fit,
  which = c("y1", "y2", "both"),
  type = c("AME", "MEM"),
  vars = NULL,
  include_intercept = FALSE,
  digits = 4,
  print_output = TRUE,
  n_cores = 1L
)
```

Arguments

<code>fit</code>	An <code>rpbnb_fit</code> object from <code>fit_rpbnb()</code> .
<code>which</code>	Which margin(s): "y1", "y2", or "both".
<code>type</code>	"AME" (average over the sample) or "MEM" (evaluated at the mean row).
<code>vars</code>	Optional variable names or indices to restrict output.
<code>include_intercept</code>	Logical; include the intercept term.
<code>digits</code>	Number of decimal places for printed output.
<code>print_output</code>	Logical; if FALSE, suppress printing.
<code>n_cores</code>	Number of worker processes for the delta-method standard-error jacobian (1 = sequential, the default). When <code>n_cores > 1</code> , the jacobian's independent per-parameter columns are dispatched across a <code>parallel::makeCluster()</code> cluster (one cluster per call, shared across every equation which computes); results are numerically identical to the sequential path. Falls back to sequential with a warning if the <code>parallel</code> package is unavailable.

Value

A data frame (single margin, invisibly) or a named list of data frames (both), each with columns Name, Estimate, StdErr, z, p, Signif, var_type.

See Also

[bnb_elasticities\(\)](#) for fixed-coefficient bnb_fit models.

Examples

```
sim <- simulate_rpbnb(n = 400,
  beta1 = c("Intercept" = 0.2, x1 = 0.4),
  beta2 = c("Intercept" = 0.1, x1 = -0.3),
  random_1 = list(x1 = list(sd = 0.5)),
  dispersion = c(m1 = 0.4, m2 = 0.5), seed = 1)
fit <- fit_rpbnb(y1 ~ x1, y2 ~ x1, data = sim$data, random_1 = "x1",
  draws = 100)
rpbnb_elasticities(fit, which = "both", type = "AME")
```

rpbnb_marginal_effects

Marginal effects for a random-parameter bivariate NB model

Description

Average marginal effects (AME) or marginal effects at the mean (MEM) for each margin of an [fit_rpbnb\(\)](#) fit, built on the Monte-Carlo integrated population mean $\mu_i = E_\beta[\exp(x_i'\beta)]$ (the same estimand as [predict_rpbnb_fit\(\)](#), reusing the fit's stored draws). Continuous effects are $\partial\mu_i/\partial x_{ij} = \text{mean}_r \text{coef}_{rj} \exp(\text{lp}_{ir})$ (the realized coefficient per draw, so random and fixed columns are handled uniformly); binary (0/1) effects are the integrated discrete difference $E[Y|x_j = 1] - E[Y|x_j = 0]$. Standard errors use a numeric delta method over the equation's mean and log-scale parameters.

Usage

```
rpbnb_marginal_effects(
  fit,
  which = c("y1", "y2", "both", "all"),
  type = c("AME", "MEM"),
  vars = NULL,
  include_intercept = FALSE,
  digits = 4,
  print_output = TRUE,
  n_cores = 1L
)
```

Arguments

fit	An rpbnb_fit object from fit_rpbnb() .
which	Which margin(s): "y1", "y2", "both", or "all".
type	"AME" (average over the sample) or "MEM" (effect at the mean row).

vars	Optional variable names or indices to restrict output.
include_intercept	Logical; include the intercept term.
digits	Number of decimal places for printed output.
print_output	Logical; if FALSE, suppress printing.
n_cores	Number of worker processes for the delta-method standard-error jacobian (1 = sequential, the default). When <code>n_cores > 1</code> , the jacobian's independent per-parameter columns are dispatched across a <code>parallel::makeCluster()</code> cluster (one cluster per call, shared across every equation which computes); results are numerically identical to the sequential path. Falls back to sequential with a warning if the <code>parallel</code> package is unavailable.

Value

A data frame (single margin, invisibly) or a named list of data frames (both/all), each with columns Name, Estimate, StdErr, z, p, Signif, var_type.

See Also

[bnb_marginal_effects\(\)](#) for fixed-coefficient `bnb_fit` models.

Examples

```
sim <- simulate_rpbnb(n = 400,
  beta1 = c("(Intercept)" = 0.2, x1 = 0.4),
  beta2 = c("(Intercept)" = 0.1, x1 = -0.3),
  random_1 = list(x1 = list(sd = 0.5)),
  dispersion = c(m1 = 0.4, m2 = 0.5), seed = 1)
fit <- fit_rpbnb(y1 ~ x1, y2 ~ x1, data = sim$data, random_1 = "x1",
  draws = 100)
rpbnb_marginal_effects(fit, which = "y1", type = "AME")
```

rpbnb_threads	<i>Number of CPU threads available for the multithreaded likelihood</i>
---------------	---

Description

Number of CPU threads available for the multithreaded likelihood

Usage

```
rpbnb_threads()
```

Value

Integer thread count reported by OpenMP (1 if OpenMP is unavailable).

simulate_bnb

*Simulate data from the Famoye/Sarmanov bivariate NB distribution***Description**

Generates paired count outcomes (y_1 , y_2) from the fixed-parameter Famoye/Sarmanov bivariate NB2 joint PMF:

Usage

```
simulate_bnb(
  n,
  beta1,
  beta2,
  dispersion = c(m1 = 0.5, m2 = 0.5),
  lambda = 0,
  covariates = NULL,
  seed = NULL
)
```

Arguments

<code>n</code>	Number of observations.
<code>beta1, beta2</code>	Named numeric vectors of coefficients for each equation; must include "(Intercept)".
<code>dispersion</code>	Named numeric <code>c(m1 = ..., m2 = ...)</code> NB2 dispersion parameters (variance = $\mu + m * \mu^2$).
<code>lambda</code>	Famoye/Sarmanov dependence parameter. Must lie within the valid bounds implied by the marginal means and dispersions.
<code>covariates</code>	Optional data frame of covariates. If NULL, standard- normal columns are generated for every non-intercept coefficient name.
<code>seed</code>	Optional random seed. If NULL (default) the RNG is left untouched and draws continue from the caller's current stream, so repeated calls yield distinct datasets; supply an integer for reproducible output.

Details

$$P(Y_1 = y_1, Y_2 = y_2) = p_1(y_1) p_2(y_2) [1 + \lambda(e^{-y_1} - c_1)(e^{-y_2} - c_2)]$$

where $c_k = E[e^{-Y_k}]$ under $NB2(\mu_k, m_k)$.

Value

A list with:

- `data` data frame with y_1 , y_2 , and covariate columns
- `mu` data frame with μ_1 , μ_2 (per-obs conditional means)
- `true` list of true parameters: `beta1`, `beta2`, `dispersion`, `lambda`
- `settings` list with `n` and `seed`
- `meta` list with `seed` and `r_version`

Examples

```
sim <- simulate_bnb(n = 500,
  beta1 = c("(Intercept)" = 0.5, x1 = 0.3),
  beta2 = c("(Intercept)" = 0.2, x1 = -0.2),
  dispersion = c(m1 = 0.4, m2 = 0.5), lambda = 0.1, seed = 1)
head(sim$data)
```

simulate_rpbnb

*Simulate data from a random-parameter bivariate NB process***Description**

Simulate data from a random-parameter bivariate NB process

Usage

```
simulate_rpbnb(
  n,
  beta1,
  beta2,
  random_1 = NULL,
  random_2 = NULL,
  dispersion = c(m1 = 0.5, m2 = 0.5),
  lambda = 0,
  covariates = NULL,
  seed = NULL
)
```

Arguments

<code>n</code>	Number of observations.
<code>beta1, beta2</code>	Named numeric vectors of fixed coefficient means per equation; must include "(Intercept)".
<code>random_1, random_2</code>	Named lists giving random coefficients. Each value is a list with <code>dist</code> (one of "normal", "lognormal", "uniform", "triangular"; default "normal"), <code>scale</code> (or <code>sd</code>) for the dispersion, and <code>sign</code> (-1/1, lognormal only). Means come from <code>beta1/beta2</code> ; for a lognormal coefficient the <code>beta</code> entry is the log-location and the realized coefficient is <code>sign * exp(log_location + scale * z)</code> .
<code>dispersion</code>	Named numeric <code>c(m1 = ..., m2 = ...)</code> NB2 dispersions.
<code>lambda</code>	Famoye dependence parameter (0 = independent margins).
<code>covariates</code>	Optional data frame of covariates; if <code>NULL</code> , standard-normal columns are generated for every non-intercept name.
<code>seed</code>	Optional random seed. If <code>NULL</code> (default) the RNG is left untouched and draws continue from the caller's current stream, so repeated calls yield distinct datasets; supply an integer for reproducible output.

Value

A list with `data` (`y1, y2, covariates`), `coef_realized` (per-obs coefficients per equation), `mu` (conditional means), `true` (parameters), `settings`, and `meta` (R/seed/timestamp passed by caller).

Examples

```
sim <- simulate_rpbnb(n = 500,
  beta1 = c("(Intercept)" = 0.2, x1 = 0.4),
  beta2 = c("(Intercept)" = 0.1, x1 = -0.3),
  random_1 = list(x1 = list(sd = 0.5)),
  dispersion = c(m1 = 0.4, m2 = 0.5), seed = 1)
head(sim$data)
```

simulate_rpbnb_copula *Simulate data from a copula RP-BNB process*

Description

Simulate data from a copula RP-BNB process

Usage

```
simulate_rpbnb_copula(
  n,
  beta1,
  beta2,
  random_1 = NULL,
  random_2 = NULL,
  dispersion = c(m1 = 0.5, m2 = 0.5),
  copula,
  covariates = NULL,
  seed = NULL
)
```

Arguments

n	Number of observations.
beta1, beta2	Named coefficient means; must include "(Intercept)".
random_1, random_2	Random-coefficient specs (see simulate_rpbnb()).
dispersion	Named c(m1=, m2=) NB2 dispersions.
copula	An copula() object giving the family and native parameter par.
covariates	Optional covariate data frame; NULL -> standard-normal columns.
seed	Optional RNG seed.

Value

list(data, mu, true, settings).

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